

Q1 Data Communication is the exchange of data between two or more devices through a Transmission medium such as cables or wireless channels

It involves a Sender, Receiver, Message, Transmission medium and Protocol

Q2 Network Topology is the physical or logical arrangement of devices and connections in a computer network

Q3 Data framing is the process of dividing a continuous stream of bits into manageable bits called frames at the data link layer

Q4 Sliding window is a flow control technique in which the sender can transmit multiple frames before receiving acknowledgments. The window moves forward (slides) as acknowledgments are received, improving transmission efficiency

Q5 Classless addressing is used to allocate IP addresses more efficiently without fixed class A, B or C boundaries. It helps in reducing IP address wastage and Routing Table size



Q6 Subnetting is the process of dividing a large IP network into smaller logical networks (called subnets). It improves IP address utilization, network performance and management.

Q7 The Transport layer provides  
• Process to process delivery  
• Segmentation and reassembly  
• Flow Control  
• Error Control  
• connection establishment and Termination  
• multiplexing and demultiplexing

Q8 Flow Control is the mechanism that prevents a fast sender from overwhelming a slow receiver. The receiver indicates how much data it can accept and the sender adjusts its transmission rate accordingly. TCP uses a receiver window for flow control.

Q9 DNS is a distributed naming system that converts domain names into IP addresses. For example, it converts a website name such as google.com into its corresponding IP address.

Q10 HTTP is an application layer used to transfer web resources between clients and servers. It follows a request-response model.